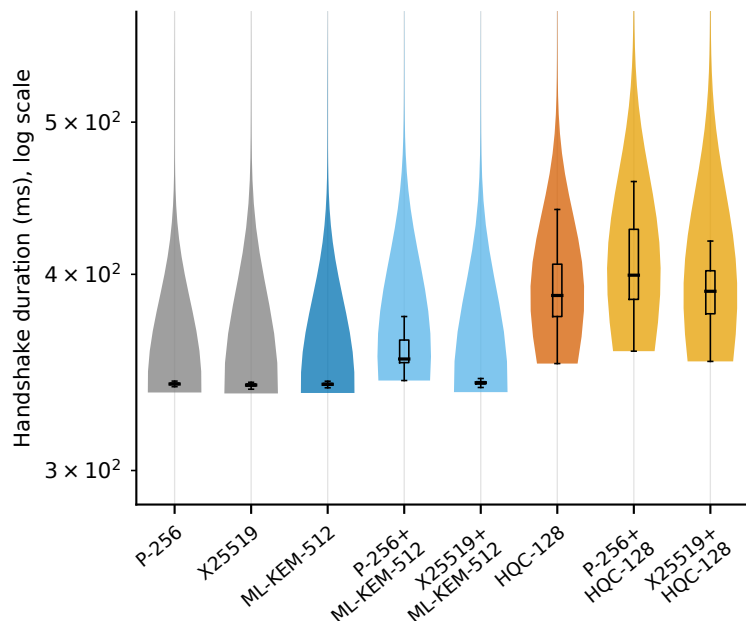
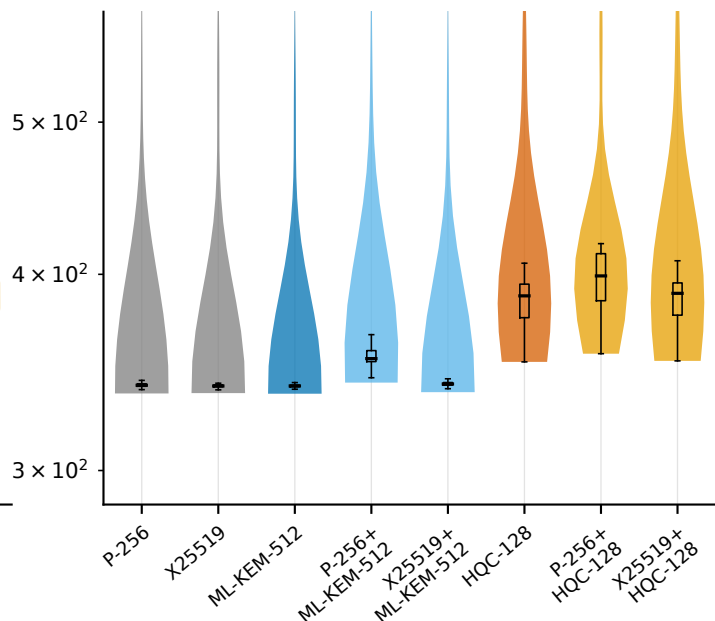


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	367.3	40.2	1.38k	381.0	51.7	1.71k
X25519	372.8	44.6	1.86k	377.7	45.3	1.92k
ML-KEM-512	371.1	40.4	1.37k	376.2	43.2	1.72k
P-256+ML-KEM-512	386.2	44.4	1.55k	395.9	49.7	2.38k
X25519+ML-KEM-512	373.0	45.1	1.88k	375.7	47.0	2.30k
HQC-128	420.5	37.2	1.74k	424.8	43.0	1.91k
P-256+HQC-128	451.8	43.1	1.88k	423.4	33.5	2.12k
X25519+HQC-128	432.5	47.7	1.77k	432.5	46.9	1.91k